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Preliminary Study

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Dynamic Liquidity Provision in Uniswap v3:
Evaluating Order-Flow Toxicity Signals

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1 Background

My interest in this topic developed from a combination of professional experience in financial consulting and a long-standing interest in financial markets. When choosing a topic for the Master's thesis, one of my main objectives was to connect this background with my academic focus in Applied Data Science and to work on a problem in which financial decision-making can be studied empirically.

Cryptocurrency markets became a natural area to explore for two reasons. First, I have been investing and trading in crypto assets for several years and was therefore already familiar with the market from a participant's perspective. Second, my Bachelor's thesis focused on traditional equity indices, where data access and data collection proved to be a practical constraint. For the Master's thesis, I wanted to work in a setting where market activity could be observed and reconstructed more directly. Public blockchain records provide such an environment and have been used to reconstruct trades, liquidity changes, and participant activity at a detailed level (Lehar & Parlour, 2025).

Within the broader crypto market, decentralized exchanges and liquidity provision became particularly interesting because they introduced an area of financial markets that I had not previously studied in depth: market making and market microstructure. This creates an opportunity to extend my financial-market knowledge while working with a setting that remains closely connected to empirical data.

Concentrated liquidity provision in Uniswap v3 was therefore selected as the initial application area for the thesis. It combines observable market activity with repeated financial allocation decisions and provides a suitable setting for studying how market information relates to subsequent outcomes.

2 Topic Definition

This study focuses on liquidity provision in automated market makers (AMMs), with concentrated liquidity as the main market design of interest. The liquidity provider is the unit of economic analysis: the problem is how capital is allocated to liquidity, how those allocations can be adjusted as market conditions change, and how the resulting outcomes should be evaluated. Uniswap v3 serves as the reference design because it makes the location and width of liquidity provision explicit provider decisions. The Thesis narrows this broader allocation problem to the conditional choice of range width. Candidate ranges are centered on the prevailing pool price, so that the primary decision concerns the degree of liquidity concentration rather than directional range placement.

2.1 Market Making and AMM-Based Trading

Traditional limit-order markets organize liquidity around quotes submitted by market makers. Liquidity suppliers can choose the prices at which they are willing to trade and may update or withdraw their quotes as market conditions change. This discretion allows them to manage the trade-off between earning compensation for supplying liquidity and being adversely selected when asset values change (Lehar & Parlour, 2025).

A Uniswap-style AMM changes this organization of liquidity. Providers do not individually set bid and ask quotes for incoming trades. Instead, they deposit assets into a liquidity pool (LP), and a deterministic pricing function determines the terms at which traders exchange against the pooled inventory. Trading fees accrue to liquidity providers, while price discovery relies partly on arbitrage between the AMM and external markets (Adams et al., 2021; Lehar & Parlour, 2025). This separation between liquidity provision and price discovery is economically important: arbitrage keeps pool prices connected to external asset values, but the resulting trades can change LP inventories at unfavorable terms when pool prices have become stale (Lehar & Parlour, 2025).

The constant-product rule underlying early versions of Uniswap provides the basic example. Abstracting from fees and changes in liquidity, reserves satisfy

$$xy = k, \tag{1}$$

where x and y denote the reserves of assets X and Y and k denotes the constant product (Adams et al., 2021; Lehar & Parlour, 2025). A trade changes both reserves and therefore moves the pool along the bonding curve. For a given trade size, greater pool depth reduces the associated price impact (Lehar & Parlour, 2025). In practice, the pool is not permanently confined to one value of k: liquidity deposits and withdrawals change pool depth, while fees retained in earlier Uniswap designs also alter the reserves (Lehar & Parlour, 2025).

Concentrated liquidity changes where that capital is available rather than abandoning the underlying automated pricing mechanism. In Uniswap v3, an LP can restrict liquidity to a bounded price range. Within that range, the position can provide greater effective liquidity for a given amount of committed capital; outside it, the position becomes inactive, stops earning fees, and is composed entirely of one of the two assets (Adams et al., 2021). The design therefore turns liquidity provision from a decision about pool participation and capital alone into a decision about the location and width of liquidity across the price space.

2.2 Liquidity-Provider Decision Problem

Concentrated liquidity gives an LP several decision variables that do not exist in the same form under full-range provision. Besides deciding how much capital to commit and to which pool, the provider must choose the location and width of the price range in which that capital is active (Adams et al., 2021; Heimbach et al., 2022). Range width creates an immediate trade-off: narrower positions use capital more intensively while they are in range, but become inactive more readily when prices move and can generate greater losses relative to holding the underlying assets (Heimbach et al., 2022).

Although both range location and width are provider-controlled variables, the empirical study isolates the range-width decision. Candidate positions will be centered on the prevailing pool price, while their width varies across a predefined set of alternatives. This restriction separates the decision about liquidity concentration from a directional forecast about future price movements.

In this study, *active liquidity management* refers to deliberate changes in these provider-controlled allocations, such as repositioning or resizing a liquidity range. This is distinct from the automatic change in token quantities inside an existing position as trades move the pool price through that range. Changes in token quantities follow mechanically from the AMM design, while repositioning or resizing a position requires a new allocation decision by the provider (Adams et al., 2021; Hasbrouck et al., 2023).

The outcome of such a decision cannot be identified with fee income alone. A liquidity position earns fees while simultaneously changing its exposure to the underlying assets as trades alter its inventory. Hasbrouck et al. (2023) therefore separate fee income from the return on the liquidity portfolio, while Heimbach et al. (2022) evaluate position value and earned fees against the value of holding the initially deposited assets. These approaches differ in their formal treatment of performance but agree that fee income by itself is insufficient to characterize the economic outcome of liquidity provision (Hasbrouck et al., 2023; Heimbach et al., 2022).

Active management also creates implementation costs. In Uniswap v3, repositioning requires additional blockchain transactions, so fixed gas costs affect both the incentive to update a position and which providers can manage liquidity frequently (Lehar et al., 2022). Performance evaluation must therefore eventually specify not only a return benchmark, but also which management costs are included. The literature review develops these alternatives before the empirical framework is fixed.

2.3 Scope and Delimitations

Empirically, the thesis focuses on liquidity provision in the WETH/USDC 0.05% Uniswap v3 pool on Ethereum because low-fee Uniswap v3 pools are characterized by high trading activity, relatively frequent liquidity adjustments, and greater exposure to adverse selection (Lehar et al., 2022). The main object of analysis is the performance of alternative liquidity-range decisions under observable market conditions. Individual LP positions and pool-level variables such as prices, trading activity, liquidity, and order-flow characteristics provide the empirical setting in which these decisions are evaluated.

Earlier AMM designs are included where they provide a necessary benchmark for understanding concentrated liquidity. Centralized markets are not a separate object of study, although external market prices are used as reference information where required for feature construction and performance measurement. The thesis is therefore concerned with liquidity provision as an investment and decision problem rather than with a broader comparison of centralized and decentralized exchange architectures.

The study does not aim to cover the wider DeFi ecosystem, design a new exchange protocol, or develop a production trading system. Lending markets, derivatives strategies, smart-contract security, and other DeFi applications fall outside the primary research objective. The analysis is also not intended to model the complete behavior of individual liquidity providers; observed LP activity is relevant as market context, while the empirical focus remains on the outcomes of alternative liquidity-allocation decisions.

The final observation period, decision frequency, prediction horizon, and exact performance measure will be specified as part of the methodological design. These choices will be guided by the information required for out-of-sample evaluation and by the level of data granularity that can be reconstructed reliably.

3 Literature Review

Building on the definitions in Section 2, this review examines liquidity provision in AMMs as both a market-design problem and an investment decision. It first considers how pricing mechanisms, fees, and arbitrage shape the incentives to supply liquidity, before examining the investment implications of concentrated liquidity. It then discusses performance measurement, observed provider behavior, and systematic liquidity management. Comparing the contributions, assumptions, and evidence across these strands provides the basis for developing a focused research question.

3.1 Market Design and the Incentives to Supply Liquidity

The economics of liquidity provision in an AMM begin with a simple trade-off: providers earn trading fees, but the same mechanism that generates those fees also exposes their inventory to arbitrage. In the constant-product setting studied by Lehar and Parlour (2025), liquidity providers do not update quotes in response to new information. Instead, outside price changes create opportunities for arbitrageurs to trade against stale pool prices. These trades help restore price alignment across venues, but they rebalance LP inventories at unfavorable terms (Lehar & Parlour, 2025).

This adverse-selection channel determines how much liquidity the market can sustain. In the model of Lehar and Parlour (2025), equilibrium pool size balances fee income against picking-off risk: stronger liquidity-trading demand supports larger pools, whereas greater exposure to informed trading reduces the amount of capital providers are willing to commit. Their empirical analysis of Uniswap v1 and v2 is consistent with this mechanism, with larger pools associated with lower token-price volatility and higher values of their proxies for uninformed trading (Lehar & Parlour, 2025).

The same economic cost is examined under a different measurement framework by Milionis et al. (2022). Whereas Lehar and Parlour (2025) model adverse selection as part of the equilibrium decision to supply liquidity, Milionis et al. (2022) isolate the losses generated by arbitrage through loss-versus-rebalancing (LVR). The two approaches answer different questions, but both locate a central cost of AMM liquidity provision in trades executed after outside prices have moved (Lehar & Parlour, 2025; Milionis et al., 2022).

Uniswap v3 adds another decision margin to this problem. Rather than spreading liquidity across the full price domain, providers can choose the price intervals in which their capital is available for trading (Adams et al., 2021). The economic question therefore shifts from only how much liquidity is supplied to how that capital is distributed across prices. Hasbrouck et al. (2023) model precisely this second margin by deriving equilibrium liquidity provision across individual price intervals. Their analysis therefore links the aggregate liquidity-supply problem of earlier AMMs to the provider-level allocation problem introduced in Section 2.2.

3.2 Concentrated Liquidity and Investment Incentives

Concentrated liquidity gives providers greater control over where their capital is deployed. As described in Section 2.1, restricting liquidity to a narrower price range increases the effective use of capital within that range. The economic question is therefore not whether concentration improves capital efficiency, but whether the resulting allocation is attractive from an investment perspective.

The economic value of that allocation is not captured by capital efficiency alone. Hasbrouck et al. (2023) show that, as the width of a price interval becomes arbitrarily small, the ex-fee payoff from concentrated liquidity approaches that of a covered-call position. A covered call combines a long position in the underlying asset with a short call option on that asset. In the setting of Hasbrouck et al. (2023), this corresponds to holding ETH while selling an ETH call with a strike price inside the liquidity interval.

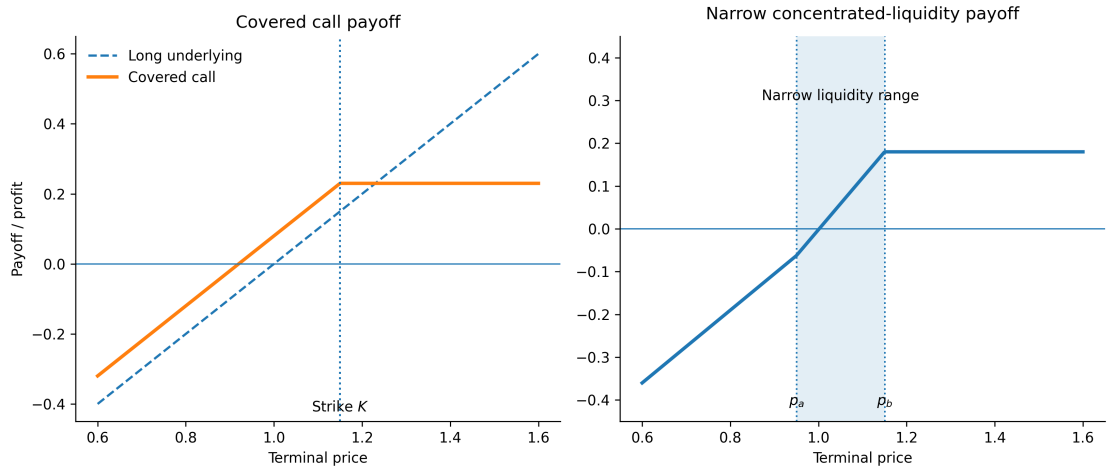


Figure 1: Covered-call payoff and stylized ex-fee payoff of a narrow concentrated-liquidity position. Own illustration based on the covered-call interpretation of concentrated liquidity in Hasbrouck et al. (2023).

The crucial difference arises at inception: the LP effectively gives up the time premium that would be received from selling the corresponding call at its market value. Positive liquidity provision therefore requires expected fee compensation for an opportunity cost that is absent from a purely technical measure of capital efficiency (Hasbrouck et al., 2023).

This result extends the fee–adverse-selection trade-off identified by Lehar and Parlour (2025). In the full-range setting, the amount of liquidity adjusts until expected compensation is sufficient to bear the costs of market making. In the concentrated-liquidity setting, Hasbrouck et al. (2023) derive the corresponding adjustment across price intervals: equilibrium liquidity decreases as the time premium associated with an interval increases (Hasbrouck et al., 2023; Lehar & Parlour, 2025). Their model also cautions against reducing the allocation problem to spot volatility alone, since option time value need not be determined by spot volatility in richer price processes (Hasbrouck et al., 2023).

The resulting trade-off is also visible in the empirical analysis of Heimbach et al. (2022). Narrow ranges increase fee-earning capacity while they remain active, but they become inactive more quickly when prices move and experience larger impermanent-loss effects. For volatile asset pairs, range width therefore affects both the revenue side and the

inventory-risk side of the LP position (Heimbach et al., 2022). Concentration should consequently be interpreted as a change in the provider's opportunity set rather than as an unconditional improvement in investment performance.

3.3 Liquidity Provider Performance and Active Management

Evaluating liquidity provision requires specifying the benchmark against which performance is measured. A common starting point is the holding benchmark used by Heimbach et al. (2022), which compares the value of a Uniswap v3 position, including earned fees, with the value of holding the initially deposited assets. Their analysis highlights a central trade-off of concentrated liquidity: narrower ranges increase capital efficiency while active, but also increase sensitivity to price movements and the likelihood of becoming inactive (Heimbach et al., 2022). Fee income alone is therefore insufficient to characterize LP performance.

A different perspective is provided by Milionis et al. (2022), who separate directional market exposure from the microstructural economics of liquidity provision. Their loss-versus-rebalancing (LVR) benchmark continuously matches the AMM's risky-asset holdings while executing the corresponding trades at external market prices. LP performance can therefore be separated into market exposure and a component consisting of fees minus losses to arbitrageurs (Milionis et al., 2022). This distinction is empirically important because their analysis of the Uniswap v2 ETH–USDC pool attributes most variation in raw LP profit and loss to market exposure rather than the market-making component.

Cartea et al. (2023) extend this benchmark perspective through predictable loss (PL), which compares LP wealth with a self-financing portfolio that tracks the same changing asset holdings and invests excess wealth in a risk-free asset. Their measure adds an opportunity-cost component to the convexity cost associated with liquidity provision. Similarly, Milionis et al. (2023) emphasize that individual LP outcomes also depend on competition for fee income. Taken together, these measures show that holding-based returns, adverse-selection losses, opportunity costs, and fee capture describe related but distinct dimensions of LP performance (Cartea et al., 2023; Milionis et al., 2022, 2023).

These economic trade-offs become particularly relevant when providers actively adjust their liquidity. Evidence from Uniswap v2 suggests relatively limited reallocation: Heimbach et al. (2021) find that most providers participate in only a small number of pools and that movements across pools are generally infrequent. More pronounced reallocations occur around liquidity-mining incentives, suggesting that providers respond when the economic incentive is sufficiently strong (Heimbach et al., 2021).

Concentrated liquidity introduces a more direct reason for adjustment because a position stops earning fees when the price leaves its range (Adams et al., 2021). Lehar et al. (2022) show that the resulting management intensity differs across providers and pools. Low-fee pools combine higher trading activity with greater adverse-selection exposure and shorter liquidity cycles, while fixed gas costs make frequent repositioning relatively more attractive for larger LPs. Active management therefore depends not only on market conditions, but also on provider-specific constraints such as capital size and transaction costs (Lehar et al., 2022).

Just-in-time liquidity illustrates the high-frequency end of this spectrum. Wan and Adams (2023) study LPs that observe a pending swap, add concentrated liquidity immediately before execution, and remove it afterwards. Although JIT demonstrates how precise information about incoming order flow can be translated into a liquidity decision, it represents only a small share of observed liquidity demand and is highly concentrated among a small number of providers (Wan & Adams, 2023). It should therefore be viewed as a specialized form of active liquidity provision rather than a representative management strategy.

Overall, the literature shows that LP outcomes depend jointly on the chosen performance benchmark, market conditions, and the costs and constraints associated with adjusting liquidity. Observing that providers react to these factors, however, does not establish that a reproducible adjustment rule improves investment outcomes. This distinction motivates the subsequent literature on systematic liquidity-management strategies.

3.4 Systematic Liquidity Management and Strategy Evaluation

Active management becomes a strategy only once the provider follows an explicit rule for choosing and updating a range. Fritsch (2021) study this distinction by comparing fixed-range and resetting strategies with both full-range liquidity provision and holding the underlying assets. Their resetting strategies separate the width of the liquidity interval from the price movement required to trigger recentering (Fritsch, 2021).

The results do not support a universally superior active rule. For ETH–USDC, some parameter combinations outperform passive liquidity in particular months, while others underperform it (Fritsch, 2021). The strategy that maximizes fee income can also perform poorly in total-value terms because the liquidity position itself loses value (Fritsch, 2021). This is the strategy-level counterpart to the distinction made by Milionis et al. (2023): capturing a larger share of fees is not equivalent to maximizing profitability when other components of LP returns deteriorate (Fritsch, 2021; Milionis et al., 2023).

Competition matters as well. Fritsch (2021) show that simulations in which an active strategy competes only against passive liquidity can substantially overstate its returns. The result reinforces the argument of Milionis et al. (2023) that individual LP outcomes depend on the behavior of competing providers, not only on aggregate pool volume or the LP's own range (Fritsch, 2021; Milionis et al., 2023). Strategy evaluation that ignores the endogenous distribution of liquidity may therefore misstate the fees that an implementable strategy could capture.

The analytical results of Cartea et al. (2023) provide a complementary view of the range-choice problem. For dynamically managed concentrated liquidity, they derive wealth dynamics for positions whose ranges move with the market price. Under their geometric-Brownian illustration with a zero risk-free rate, the magnitude of predictable loss increases with volatility and falls as the position spread becomes wider (Cartea et al., 2023). The result gives an economic reason for adapting concentration to market conditions, but it does not by itself specify an implementable rule for selecting the range from observable data.

The empirical and analytical approaches therefore expose different parts of the same problem. Fritsch shows that the ranking of simple range strategies changes with market conditions and that the best-performing parameters are identified only after the realized price path is known (Fritsch, 2021). Cartea et al. explain why range width changes the underlying loss exposure (Cartea et al., 2023). Together with the evidence on heterogeneous adjustment costs in Lehar et al. (2022), these results make a fixed notion of "active management" too coarse: the relevant decision is which adjustment to make under the market conditions observable at that point in time (Cartea et al., 2023; Fritsch, 2021; Lehar et al., 2022).

3.5 Market Information and Adaptive Liquidity Decisions

The strategy evidence changes the problem from one of parameter selection to one of information. If the relative performance of different ranges varies across market conditions, an active LP needs some basis for deciding which condition currently applies. Fritsch (2021) identify the best strategy parameters only after observing the realized period, while Heimbach et al. (2022) show that variables such as volatility affect fee opportunities, out-of-range risk, and impermanent loss simultaneously (Fritsch, 2021; Heimbach et al., 2022). Historical state dependence is therefore not the same as an ex-ante decision rule.

Adverse selection is one candidate source of state information. Milionis et al. (2022) separate arbitrage losses from directional market exposure and thereby isolate a component of LP performance that is specific to trading against better-informed counter-

parties. Milionis et al. (2023) place this measure alongside LP competition, treating LVR as a measure of flow toxicity and FLAIR as a measure of the fee return captured relative to competing liquidity (Milionis et al., 2022, 2023). This decomposition is useful for diagnosing realized performance, but FLAIR itself is constructed as an ex-post measure (Milionis et al., 2023). A signal that describes harmful order flow after it occurred is not automatically a signal that can improve the next allocation decision.

The behavior documented by Lehar et al. (2022) is consistent with such state dependence without resolving this prediction problem. Providers in low-fee pools face more trading activity, higher adverse-selection costs, and shorter liquidity cycles, while gas costs and provider size affect how frequently adjustment is economically viable (Lehar et al., 2022). Observable market conditions therefore coincide with different management behavior, but the empirical relationship does not itself specify which range an LP should choose next.

JIT liquidity provides a useful limiting case because the information problem is unusually small. The LP observes a pending transaction before deciding where to provide liquidity, so the size and location of the next trade are already known when the position is created (Wan & Adams, 2023). The strategy demonstrates that sufficiently precise information can be translated directly into a liquidity decision. Its rarity and concentration among very few providers, however, make it a specialized case rather than evidence that ordinary market-state variables are equally informative (Wan & Adams, 2023).

A recent non-peer-reviewed study by Likassi (2026) examines whether additional toxicity information improves adaptive liquidity management. The author augments a PPO-based Uniswap v3 strategy with several proxies for recent flow toxicity and compares the resulting agents with a toxicity-blind adaptive baseline. While all PPO configurations outperform a passive fixed-range benchmark, none of the toxicity-aware specifications achieves statistically significant superiority in mean return over the toxicity-blind agent (Likassi, 2026). The results provide exploratory evidence that some toxicity signals may instead be more informative for downside risk, but the paper has not yet undergone peer review and its findings should therefore be interpreted cautiously.

The literature therefore narrows the adaptive-management problem considerably. The open issue is no longer simply whether market conditions matter for concentrated-liquidity outcomes; both empirical and analytical work already show that they do (Cartea et al., 2023; Fritsch, 2021; Heimbach et al., 2022; Lehar et al., 2022). The more demanding question is whether information observable before an allocation decision contains incremental and sufficiently robust information to improve that decision relative to a suitable state-aware benchmark. Recent exploratory work suggests that flow-related signals may be relevant, but stronger evidence is required before drawing

conclusions about their predictive value (Likassi, 2026; Milionis et al., 2023).

4 Main Research Question

Across the literature, liquidity-management outcomes in concentrated-liquidity AMMs vary with market conditions. However, signals that explain realized performance are not necessarily useful for decisions made in real time. Recent toxicity-aware reinforcement-learning studies address this issue by incorporating market-state information into adaptive liquidity management, but do not establish a consistent improvement in mean returns across settings.

Primary Research Question

To what extent can market information available at the time of decision support the out-of-sample selection of liquidity-range width in Uniswap v3?

Sub-Questions

1. Which observable market-state variables contain predictive information about the subsequent relative performance of alternative liquidity-range decisions?
2. Do order-flow toxicity measures provide incremental predictive value beyond standard market-state variables such as volatility and trading activity?
3. Can a range-width selection rule based on these predictions improve out-of-sample LP outcomes relative to static or simple rule-based benchmarks after accounting for implementation costs?

The sub-questions follow the sequence from prediction to decision. First, they identify which market variables help explain subsequent strategy performance. Second, they test whether order-flow toxicity adds information beyond standard market variables. Finally, they examine whether these signals can improve out-of-sample range selection relative to simpler benchmarks.

5 Methodology

The thesis design follows directly from the distinction developed in the literature review between retrospective performance measurement and prospective liquidity-allocation decisions. The objective is not to explain the historical choices of individual liquidity providers, but to evaluate whether information observable at a decision point can help distinguish between alternative liquidity ranges before their subsequent outcomes are

known. The analysis therefore combines counterfactual strategy evaluation with time-series prediction and out-of-sample testing.

5.1 Research Design

At each decision time t, a hypothetical liquidity provider observes the market information available up to that point and chooses from a predefined set of alternative range widths $A = \{a_1, \ldots, a_K\}$. Each candidate position is centered on the prevailing pool price and differs only in its width. Range boundaries will be mapped to valid Uniswap v3 ticks, and the same action set will be used throughout the main analysis.

For every decision time and every candidate range, the subsequent position will be reconstructed under the same realized market path over a fixed horizon H. The resulting data structure therefore contains multiple counterfactual outcomes for each observed market state. In general form,

$$Y_{t,a}^{(H)} = \text{performance of range } a \text{ selected at } t \text{ over } [t, t+H], \tag{2}$$

where $Y_{t,a}^{(H)}$ denotes the economic outcome associated with action a at decision time t. The primary specification will use a common initial capital amount and a fixed decision horizon for all alternatives. The final horizon and range grid will be fixed before the final predictive analysis and will not be selected on the basis of holdout performance. Alternative horizons and range definitions will subsequently be used as robustness checks.

The counterfactual positions will be evaluated using the realized sequence of Uniswap swaps and liquidity changes. Rather than assuming that all trading activity within an aggregation interval occurred at one representative price, the underlying on-chain observations will be retained at event level where feasible. Liquidity additions and removals will be processed chronologically, and swaps crossing multiple ticks will be evaluated against the liquidity available along the realized price path. This avoids the hourly closing-price approximation used in earlier strategy simulations such as Fritsch (2021).

The hypothetical provider will be treated as a marginal liquidity supplier whose capital is small relative to the pool. The realized market path and the liquidity supplied by other participants are therefore taken as given. This assumption makes the counterfactual outcomes interpretable as the performance that a sufficiently small provider would have obtained under the observed market conditions; it does not imply that large hypothetical positions could have been added without affecting market prices or other liquidity providers.

The raw data will be transformed into a reproducible analytical dataset before the strategy simulation. Duplicate blockchain events will be identified using transaction and log identifiers, failed transactions will be excluded, token decimals and price orientation will be standardized, and all timestamps will be converted to a common UTC time basis. Pool-state variables derived from the event sequence will be checked for internal consistency with observed Uniswap swap states. External price observations will be aligned using only information available at or before the relevant decision time. Missing reference-market observations will not be interpolated across material gaps. Genuine periods of unusually high volatility, volume, or transaction costs will remain in the sample, since these observations are economically relevant to the research question. Strongly skewed variables may instead be transformed using logarithms or robust scaling.

5.2 Analytical Framework

Raw LP returns combine liquidity-provision performance with directional exposure to the underlying assets. A Uniswap v3 position changes its token composition as the market price moves through the selected range. Consequently, raw changes in position value partly reflect directional exposure to the underlying assets rather than the quality of the liquidity decision itself.

Following the decomposition proposed by Milionis et al. (2022), the primary performance measure will therefore compare the LP position with a rebalancing portfolio that follows the same risky-asset exposure but trades at the external reference price. For a candidate range a, the net LP profit over the decision horizon can be written as

$$\Pi_{t,a}^{LP} = V_{t+H,a}^{LP} + F_{t,a} - V_{t,a}^{LP} - C_{t,a}, \tag{3}$$

where V^{LP} denotes the marked-to-market value of the liquidity position, $F_{t,a}$ denotes fees earned during the interval, and $C_{t,a}$ contains the implementation costs associated with liquidity management. Fees will be accumulated separately from the position, consistent with the non-compounding treatment of fees in Uniswap v3.

Let $x_{a,\tau}$ denote the amount of the risky asset held by the counterfactual LP position during interval τ, and let P_{τ}^{ref} denote its external market price. A discrete approximation to the corresponding rebalancing-portfolio profit is

$$\Pi_{t,a}^{RB} = \sum_{\tau=t}^{t+H-1} x_{a,\tau} \left(P_{\tau+1}^{ref} - P_{\tau}^{ref} \right). \tag{4}$$

The primary target is then defined as

$$Y_{t,a} = \Pi^{LP}_{t,a} - \Pi^{RB}_{t,a}. \tag{5}$$

This measure removes the principal directional market-risk component and places the empirical focus on the part of performance more closely associated with fee income, adverse selection, range placement, and implementation costs. This distinction is relevant because Milionis et al. (2022) show that raw LP profit and loss can be dominated by the market exposure of the underlying assets.

The market-risk-adjusted outcome will be complemented by economically different secondary measures. These include total net position return and performance relative to holding the assets initially deposited into the position. Reporting the measures separately makes it possible to distinguish whether a range decision improves the economics of liquidity provision from whether the overall investment outperforms a passive asset-holding benchmark.

Implementation costs will be incorporated explicitly. Repositioning a Uniswap v3 position requires blockchain transactions, and fixed gas costs affect the attractiveness of frequent liquidity adjustments (Lehar et al., 2022). Historical transaction costs will therefore be used where they can be reconstructed reliably. The main strategy comparison will use identical decision times, capital assumptions, and cost treatment across competing range-selection rules, so that differences in performance are not mechanically generated by different rebalancing frequencies.

5.3 Statistical and Econometric Methods

Before estimating predictive relationships, the time-series properties of the market variables and counterfactual outcomes will be examined. Price levels will generally be replaced by returns or relative-price measures, while persistent quantity variables will be transformed where necessary. Autocorrelation functions and Ljung–Box tests will be used to assess serial dependence in relevant series. Augmented Dickey–Fuller and KPSS tests will be applied to continuous variables for which stochastic trends remain a concern after transformation. These tests are intended as diagnostics for the econometric specification rather than as mechanical screening rules for the predictive models.

The first research sub-question will initially be examined with an interpretable panel-style regression in which observations are indexed by decision time and candidate range. Range fixed effects capture systematic differences between narrower and wider positions, while interactions between the range alternatives and lagged market-state variables allow the relation between market conditions and subsequent performance to vary across actions. A generic specification is

$$Y_{t,a} = \alpha_a + \boldsymbol{\beta}'\mathbf{X}_t + \boldsymbol{\gamma}'\left(\mathbf{X}_t \times \mathbf{A}_a\right) + \varepsilon_{t,a}, \tag{6}$$

where $\mathbf{X}_t$ contains information observable at decision time t and $\mathbf{A}_a$ identifies the candidate range. Standard errors will account for temporal dependence and the fact that alternative range outcomes observed at the same decision time are not independent. Where appropriate, Driscoll–Kraay or heteroskedasticity- and autocorrelation-consistent inference will be used.

The second research sub-question will be addressed through nested information sets. A baseline specification will contain conventional market-state variables such as recent returns, realized volatility, trading volume, swap activity, liquidity depth, and transaction costs. A second specification will add the predefined order-flow toxicity variables. The relevant comparison is therefore not whether a toxicity measure is correlated with subsequent LP outcomes in isolation, but whether it improves prediction conditional on information that an LP could already obtain from standard market variables.

Statistical evidence will be reported together with estimated effect sizes and confidence intervals. For out-of-sample model comparisons, the time series of forecast-loss or decision-loss differences will be evaluated using dependence-aware procedures such as moving-block bootstrap confidence intervals. Diebold–Mariano-type comparisons with autocorrelation-robust variance estimates may be used as a complementary test. Where several toxicity variables or model specifications are tested individually, multiplicity adjustments such as the Holm procedure will be applied to the corresponding family of hypothesis tests.

5.4 Data Science / Machine Learning Component

Prediction becomes relevant once the task is to rank feasible ranges using only information available at time t. At decision time t, the task is to estimate the subsequent performance of each feasible range using only the state information available at that time. The general prediction function is

$$\widehat{m}(\mathbf{X}_t, a) \approx \mathbb{E}\left[Y_{t,a} \mid \mathbf{X}_t, a\right]. \tag{7}$$

The estimated range-selection rule then chooses

$$\widehat{a}_t = \arg\max_{a \in \mathcal{A}} \widehat{m}(\mathbf{X}_t, a). \tag{8}$$

This formulation allows the predictive value of the market state to be studied separately from the optimization procedure. It also avoids assuming in advance that a

reinforcement-learning architecture is required. Given the tabular and time-series nature of the expected dataset, the main comparison will use a regularized linear model as an interpretable predictive baseline and a gradient-boosted tree model to capture nonlinearities and interactions. More complex models will only be considered if they provide a clear methodological advantage.

The feature set will be divided before model estimation into conventional market-state and toxicity-related variables. All rolling variables will be constructed from windows ending at or before the decision point; observations from the prediction horizon will never enter the feature set. Model preprocessing, variable scaling, feature selection, and hyperparameter tuning will be estimated exclusively on the corresponding training period.

Model evaluation will preserve the temporal ordering of the data. Random train-test splits will not be used. The sample will instead be divided chronologically into a development period and a final contiguous holdout period. Within the development period, model selection will use expanding-window or rolling-window validation. Where feature windows or outcome horizons overlap fold boundaries, observations will be purged and an embargo corresponding to the relevant horizon will be introduced to prevent information leakage.

Conventional forecast metrics such as mean absolute error and out-of-sample R^2 will be reported, but they are not sufficient for the economic decision problem. The analysis will also measure whether the predicted ordering of the candidate ranges matches their realized ordering. A central decision metric will be regret,

$$\text{Regret}_t = \max_{a \in \mathcal{A}} Y_{t,a} - Y_{t,\widehat{a}_t}, \tag{9}$$

which measures the performance forgone because the prediction rule did not select the ex-post best available range. The economic value of the resulting strategy will then be evaluated from its realized sequence of net out-of-sample outcomes.

5.5 Robustness and Validation

Robustness will be assessed along several predefined dimensions. Temporal stability will be examined across walk-forward evaluation periods and market regimes defined ex ante using realized volatility. The analysis will also be repeated using alternative decision horizons, candidate range grids, and performance measures, including market-risk-adjusted LP performance, total net LP return, and performance relative to holding the initially deposited assets.

Sensitivity to implementation assumptions will be evaluated by varying transaction

costs, reference capital, execution delay, and the construction of the external reference price. Model dependence will be assessed by comparing the main predictive model classes and by evaluating otherwise identical specifications with and without toxicity-related variables.

Falsification tests will examine whether apparent predictability reflects economically meaningful information or only temporal dependence in the data. Selected toxicity variables will therefore be shifted in time or permuted within appropriate time blocks and compared with correctly aligned signals.

Final strategy performance will be evaluated on a contiguous holdout period that is excluded from feature selection, model tuning, and specification choice. Benchmark strategies will use the same decision times, capital assumptions, and transaction-cost treatment as the predictive strategy, thereby isolating the value of range selection from differences in trading frequency.

Finally, the data-reconstruction pipeline will be validated independently of the predictive results. Reconstructed pool prices, ticks, and active liquidity will be checked against observable Uniswap states and, where necessary, against an alternative public blockchain-data source. Extraction queries, preprocessing rules, model configurations, and random seeds will be retained to support reproducibility.

6 Underlying Data

All required data are available from public on-chain and centralized-exchange sources. The data architecture combines on-chain information from Uniswap v3 with high-frequency price data from a centralized exchange. This allows the state of the liquidity pool and its incoming order flow to be observed jointly with an external reference price.

6.1 Primary Data Sources

On-chain observations will primarily be obtained from Dune Analytics. Dune provides decoded Ethereum blockchain data that can be queried using SQL and therefore allows interactions with Uniswap v3 smart contracts to be extracted without operating a full Ethereum node. The analysis will focus on the WETH/USDC 0.05% pool on Ethereum mainnet. Relevant observations include swap transactions as well as liquidity additions and removals. Depending on the subsequent feature construction, additional Ethereum transaction information can be joined through transaction hashes to obtain block information and transaction costs associated with liquidity management.

Centralized-exchange data will complement the on-chain observations. Public Binance spot-market data will be used as an external reference for the market price of ETH.

Historical trades and candlestick data are publicly available and can be downloaded without a commercial data subscription. These observations provide the basis for constructing market returns, realized volatility, and deviations between the contemporaneous Uniswap and external market price.

To reduce dependence on a single data provider, public Ethereum data accessible through Google BigQuery will serve as an alternative source for validating or reconstructing critical on-chain observations. The BigQuery Sandbox provides access to public datasets within predefined free usage limits. Consequently, the feasibility of the proposed analysis does not depend on continued access to a paid blockchain-data service.

6.2 Data Coverage and Granularity

WETH/USDC 0.05% on Ethereum forms the primary on-chain sample, with Binance ETH/USDT serving as the external reference market. The final sample period will be determined after consultation with the thesis supervisors and fixed prior to the start of the masters thesis.

Uniswap observations will initially be retained at their native transaction and event level, preserving block timestamps, block numbers, and transaction ordering. This granularity allows short-lived liquidity changes and the sequence of swaps and liquidity updates to be reconstructed without prematurely discarding microstructure information.

External reference-market data from Binance will be collected at one-minute frequency. Where required for specific microstructure measures, more granular trade-level data may additionally be used. Following synchronization and feature construction, the on-chain and off-chain data can be aggregated to the decision frequency employed in the empirical strategy evaluation. The final decision frequency will be specified as part of the methodological design.

6.3 Data Limitations

Several limitations remain. First, Dune provides a processed representation of public Ethereum data, and its schemas or free-access conditions may change. Extraction queries will therefore be documented and critical variables cross-validated against public Ethereum data where necessary.

Second, Binance serves only as a proxy for the external market price. Differences in market structure, timestamps, and the use of USDT rather than USDC may introduce measurement error when synchronizing CEX and DEX prices.

Third, reconstructing Uniswap v3 liquidity requires chronological processing, as liquidity varies across ticks and individual swaps may cross multiple price ranges.

Finally, focusing on a single liquid WETH/USDC pool improves data quality and comparability but limits generalizability to other pairs, fee tiers, and less liquid markets. Blockchain addresses will be treated solely as pseudonymous technical identifiers, with no attempt to identify individual users.

7 Technology, Software, and Applications

Python 3.11 will serve as the programming language. Development will take place in Visual Studio Code. Google Colab may additionally be used where cloud-based computing resources are useful, particularly for computationally intensive model training.

7.1 Key Libraries and Frameworks

Data processing and numerical analysis will rely on libraries such as `pandas`, `NumPy`, `SciPy`, and `scikit-learn`. `Matplotlib` will be used for visualisation. If needed, additional libraries such as `statsmodels`, `PyTorch`, or reinforcement-learning frameworks may be incorporated. Data-access libraries will be selected according to the APIs and data sources used in the final implementation.

7.2 Data Infrastructure

Extracted data will be stored locally in Apache Parquet format, which provides efficient compressed storage for transaction-level datasets. If required, lightweight analytical tools such as `DuckDB` may be used for querying Parquet files.

7.3 Reproducibility

Source code, extraction queries, configurations, random seeds, configuration and documentation will be maintained in a dedicated GitHub repository. A virtual environment (`venv`) will be used to isolate dependencies, with package versions documented in a `requirements.txt` file.

8 Annotated Disposition

The following outlines the anticipated structure of the Master's Thesis.

1. **Introduction**
   Introduces the motivation, research problem, research questions, and expected

contribution of the thesis.

2. **Background and Institutional Setting**
   Explains the AMM and Uniswap v3 mechanisms required to understand concentrated liquidity provision and its main economic risks.
3. **Literature Review**
   Reviews prior research on LP economics, performance measurement, active liquidity management, and market information, and derives the research gap.
4. **Data and Methodology**
   Describes the data sources, feature construction, predictive framework, benchmark range decisions, and out-of-sample evaluation design.
5. **Empirical Analysis and Results**
   Evaluates the predictive models and range-selection rules and compares their performance with relevant benchmarks.
6. **Discussion and Robustness**
   Interprets the findings, assesses robustness and limitations, and relates the results to the existing literature.
7. **Conclusion**
   Summarizes the main findings, contribution, limitations, and directions for future research.

9 Project Risks

Several implementation and methodological risks could affect the project.

Table 1: *Project Risks and Mitigation Measures*

Risk	**Likelihood**	**Impact**	**Mitigation**
Data access restrictions	Low–Medium	High	Use only public or free data sources and maintain alternative access routes.
Data quality and synchronization	Medium	High	Validate timestamps, price alignment, and aggregation procedures across venues.
Complexity of Uniswap v3 reconstruction	Medium	High	Limit reconstruction to variables required for the final empirical analysis.
Methodological complexity	Medium	High	Maintain simpler rule-based or machine-learning approaches as fallback methods.

Table 1 (continued)

Risk	Likelihood	Impact	Mitigation
Computational requirements	Low–Medium	Medium	Use efficient data formats and reduce model complexity if required.
Scope expansion	Medium	High	Keep the core analysis focused on one primary WETH/USDC pool.
Time constraints	Medium	High	Establish a working baseline and evaluation pipeline early in the project.

Note. Likelihood and impact are assessed qualitatively as low, medium, or high based on the expected probability of occurrence and potential effect on successful project completion.

10 Work and Research Plan

Table 2: *Overview of Research Phases*

Phase	Period	Key Deliverables
Preliminary Study	Until 23 Oct. 2026	Preliminary Study; validated data access and pilot extraction.
Data acquisition	26 Oct.–20 Nov. 2026	Complete raw dataset and documented extraction pipeline.
Market reconstruction	23 Nov.–18 Dec. 2026	Validated reconstruction of relevant Uniswap v3 pool states.
Data preparation and exploratory analysis	4–22 Jan. 2027	Clean analytical dataset, feature definitions, and time-series diagnostics.
Counterfactual strategy simulation	25 Jan.–19 Feb. 2027	Validated LP simulator, performance targets, and benchmark strategies.
Statistical and predictive modelling	22 Feb.–26 Mar. 2027	Econometric analysis and predictive models.
Out-of-sample evaluation	29 Mar.–16 Apr. 2027	Holdout evaluation, statistical tests, and robustness checks.

Table 2 (continued)

Phase	**Period**	**Key Deliverables**
Results and thesis writing	19 Apr.–7 May 2027	Completed empirical analysis and full thesis draft.
Revision and finalization	10–28 May 2027	Final revisions, reproducibility checks, and submission.

Note. The schedule is structured so that the main empirical analysis is completed before the final writing and revision phase. Dates indicate planned completion periods and may be adjusted within phases without changing the final submission deadline.

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